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COURSE CODE: DJS23ICPC501 **DATE: 16/10/2026**

COURSE NAME: Computer Networks **CLASS: T. Y. B.Tech**

Experiment No. 9

CO/LO: Implement Transport Layer Protocols.

Aim: To implement sockets using UD

Description of Experiment:

UDP stands for User Datagram Protocol. UDP is a connection-less and unreliable protocol of transport layer. UDP does not establish a connection between two communicating parties before transmitting the data. Following are the steps given that is to be followed for establishing UDP socket connection on the client-side Use socket() function for creating socket; recvfrom() and sendto() functions are used for transmitting data between two communicating parties.

CODE:

Server Side

```
import socket

server_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM) print("UDP
socket created.")

print("Socket bound to port 8080.")

client_addr = server_socket.recvfrom(1024)
print("Received from client:", data.decode())
```

```
  response = "Hello from server!"
  server_socket.sendto(response.encode(), client_addr)
  print("Response sent to client:", client_addr)

server_socket.close()
```

Client Side

```
import socket

client_socket = socket.socket(socket.AF_INET, socket.SOCK_DGRAM)

server_ip = input("Enter server IP address: ") server_address
= (server_ip, 8080)

msg = "Hello Server!"
client_socket.sendto(msg.encode(), server_address)
print("Message sent to server.")

addr = client_socket.recvfrom(1024)
print("Server says:", data.decode())

client_socket.close()
```

```
PS D:\College Work\Sem 5\CN> python exp9s.py
UDP socket created.
Socket bound to port 8080.
Received from client: Hello Server!
Response sent to client: ('127.0.0.1', 51321)
PS D:\College Work\Sem 5\CN>
```

```
PS D:\College Work\Sem 5\CN> python exp9.py
Enter server IP address: 127.0.0.1
Message sent to server.
Server says: Hello from server!
PS D:\College Work\Sem 5\CN>
```

Conclusion:

Thus, we successfully implemented UDP socket programming using Python to establish communication between a client and a server. The experiment helped us understand how the socket(), sendto(), and recvfrom() functions are used to send and receive messages. We also learned how UDP enables simple, connectionless communication between applications over a network.